

Workshop · Half day · 180 minutes

RAG that survives your actual documents

You leave with twenty questions your own index cannot answer yet, and the named reason each one failed.

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How this runs

0:00	Why the demo worked Twelve clean documents, and then the real folder arrived
0:40	Filing, not searching You audit your own corpus and find what it cannot answer
1:35	Measuring the retrieval, not the answer Twenty questions, and the document that should win each one
2:25	Grounding and the bill Citations, refusals, and what the whole thing costs a month

Four true things. One wrong answer.

1

The document was there

Indexed on Monday.
Nothing failed, nothing
timed out.

2

The model was the good one

The same model that nailed
every question in the demo.

3

The prompt was careful

Written by the person who
knows the policy best.

4

The answer was confident

Fluent, structured, with a
citation at the end of it.

It cited the 2019 Christmas party memo.

Nothing broke. The right document was in the corpus and was never retrieved.

01

Your retrieval problem is a filing problem

By the end of this block you can name which of six failure modes your own system has, and what each one costs you.

Every newsroom had a clippings desk

1

Somebody filed it

Every story was cut out, dated, and put in a labelled folder the same afternoon.

2

Somebody pulled it

A reporter said a name and got back a thin stack, not the whole cabinet.

3

Somebody printed it

With a byline and a dateline, so a reader could go and check.

Nobody ever fixed a bad clippings desk by hiring a better reporter.

Retrieval is three jobs wearing one name

INDEX

Cut and file

Split documents into chunks, turn each into a vector, store it with whatever you kept about where it came from.

SEARCH

Pull the stack

Turn the question into a vector, return the k nearest chunks. Usually k is 5. Nobody chose 5.

GENERATE

Write it up

Hand those chunks to the model with the question. It answers from what it was handed, and only that.

The model never sees your corpus. It sees five paragraphs somebody else picked.

Six ways it breaks, and who pays

FAILURE	WHAT YOU SEE	WHAT IT COSTS
Not indexed	Confident answer, wrong source	Trust, once
Chunked mid sentence	Half a rule, stated as whole	A wrong decision
Similar, not relevant	Five chunks on the theme	Rework, every query
Stale index	Last quarter's policy	The compliance call
No provenance	Nobody can verify it	Adoption stalls
Scanned page, no text	Silence on the key document	The whole use case

Same question. Same index. Two answers.

ASKED

How many days of parental leave do we offer in Germany?

ANSWERED

Sixteen days, per the attached policy. The retrieved chunk was an internal events memo that used the phrase "family days" twice.

It did not hallucinate. It answered accurately from the wrong paragraph.

Similar and relevant are not the same word

WHAT THE VECTOR KNOWS

- These two paragraphs use the same vocabulary
- They sit near each other in meaning space
- One scores 0.81, the other 0.79

WHAT THE VECTOR CANNOT KNOW

- Which one is current
- Which one applies to Germany
- Which one is the policy and which is a summary of it

Everything in the right hand column is metadata you either kept or threw away at indexing time.

Nobody opens the folder before they index it

1

Count the scans

Pages with no extractable text return nothing and report no error. Grep for it.

2

Count the duplicates

Four near identical versions of one policy will split your retrieval four ways.

3

Count the undated

A document with no date in it cannot lose to a newer one. It will win forever.

One hour with the folder open beats one week of tuning.

You will find the reason for three of your six failure modes before you write any code.

02

A chunk is a paragraph with an address

By the end of this block you have audited your own corpus and can say which documents it will never answer from.

A chunk has to stand up on its own

1

It says what it is about

"As stated above, this does not apply"
retrieves perfectly and means
nothing.

2

It carries its address

Document, section, version, date.
Attached to the chunk, not to the file.

3

It answers one thing

If it spans two rules, it will be
retrieved for both and be half right
for each.

The test: if a new joiner could not answer from the chunk alone, neither can the model.

Split on structure, not on character count

800 CHARACTERS, 100 OVERLAP

- Two lines to write, works on anything
- Cuts tables in half, every time
- Splits a numbered rule from its number
- Your default today, almost certainly

HEADING, SECTION, TABLE

- Needs a parser per document type
- Keeps a rule with its exception
- Gives you the heading path for free
- Two days of work, once

Structure aware chunking is the single change with the largest effect per hour spent.

Five fields you wish you had kept

FIELD	WHAT IT BUYS YOU	COST TO ADD
Effective date	Newest version wins	One regex
Jurisdiction	Filter before you search	Folder path
Document type	Policy beats summary	Filename rule
Heading path	The chunk knows its topic	Free, if parsed
Source URL	The reader can verify	Free, at ingest

Nobody gets stuck at the door

A · RUN IT

Your own corpus

You have a folder and a terminal.
Point the script at your real
documents.

B · LITE

Any chat window

Paste five of your real documents in
and run the same audit by hand.
Same lesson, smaller n.

C · WATCH

Follow on screen

Use the sample corpus we hand over
and run it at your desk tonight.

CHECKPOINT

Call out your letter. A, B or C.

Audit the folder before you trust the index

TYPE THIS

```
For each file: name, page count, is  
any text extractable, newest date  
inside, and does another file repeat  
its first paragraph.
```

One rule: point it at the real folder, not a cleaned copy. Ask for one row per file and no summary. Notice the rows with an empty text column and the rows with no date.

CHECKPOINT How many of your files came back with no extractable text? Say the number out loud.

03

You cannot fix what you have never scored

By the end of this block you have twenty questions, the document that should win each one, and a number you can watch move.

Recall at k in one sentence

1

Write the question

A real one, from a real person, phrased the way they actually phrase it.

2

Name the document

The one a knowledgeable colleague would hand over. One line, by hand.

3

Count the hits

Of twenty questions, in how many did that document appear in the top five? That is recall at 5.

It takes ninety minutes to build and it is the only number in this deck that predicts anything.

You are not debugging an answer. You are debugging a retrieval.

Every hour spent rewriting the prompt to fix a wrong fact is an hour spent on the wrong layer.
Score the retrieval first. Then the answer is a separate, smaller problem.

Twenty rows tell you more than a dashboard

QUESTION	SHOULD RETRIEVE	TOP 5
Parental leave in Germany	HR policy 2026, sec 4	No
Notice period, 3 year staff	Contract template v7	Yes
Who signs a 40k invoice	Approval matrix 2026	Yes
Expensing a home desk	Expense policy, annex B	No
Data retention, CVs	Retention schedule 2025	No

Score the stack before you score the story

RETRIEVAL EVAL

- Did the right document come back
- Answer is yes or no, no judgement
- Runs in seconds, costs nothing
- Fix here first, always

ANSWER EVAL

- Given the right document, was the answer right
- Needs a human or a grading model
- Runs in minutes, costs real money
- Meaningless until retrieval is fixed

Run them together and every failure looks like a model problem.

Where the twenty questions come from

DO

- Take them from last month's support tickets
- Keep the original phrasing, typos included
- Include five you know it gets wrong
- Name the document before you run anything

DON'T

- Write questions you already know it answers
- Clean the wording into policy language
- Let the system pick the expected document
- Stop at five because twenty felt long

If your set has no failures in it, you wrote the set to pass.

Build the set and get your first number

TYPE THIS

```
Run these twenty questions. Return  
the top five chunk sources for each,  
and mark whether the expected  
document is among them.
```

One rule: do not change the questions after you see the score. Report the count, then only the failures, and notice whether they share a document type or a date range.

CHECKPOINT What is your recall at 5, out of 20? First number, not the tuned one.

04

An answer nobody can check is not an answer

By the end of this block you can cite, refuse, and say what the whole thing costs a month.

The byline and the dateline do the trust work

1

Cite the chunk, not the file

"HR policy 2026, section 4" sends them to the paragraph. A filename sends them on a search.

2

Show the date

A reader who sees 2019 next to the answer catches the error you never will.

3

Link it, do not describe it

If verifying costs more than re-asking a colleague, nobody verifies and nobody adopts.

Citations are not decoration. They are how the room finds your retrieval bugs for free.

Refusing is a product feature, not a failure

1

Set a floor

Below a similarity score you pick, return nothing and say so. Pick it from your own eval rows.

2

Say what is missing

"No current policy covers this in Germany" is useful. "I cannot help with that" is not.

3

Route the refusal

Every refusal is a question your corpus should answer and does not. Log them and read them weekly.

A system that never refuses is a system that always answers, including when it should not.

Your refusal log is the best backlog you will ever get, and it writes itself.

Five levers, and what each one moves

LEVER	EFFECT ON COST	EFFECT ON QUALITY
Smaller k	Down, linearly	Down, if recall is weak
Re-rank top 20 to 5	Up a little	Up a lot
Cache frequent questions	Down hard	Flat, if invalidated
Cheaper embedding model	Down at ingest only	Down, measurably
Cheaper answer model	Down hard	Flat, once grounded

Do the sum before the pilot, not after

THE INPUTS YOU ALREADY KNOW

- Questions per day, from your ticket log
- Chunks retrieved per question
- Tokens per chunk, roughly 300
- Re-index frequency, per month

THE NUMBER THAT DECIDES IT

- Monthly answer cost, from the four above
- Monthly ingest cost, usually a rounding error
- Minutes saved per question, measured not assumed
- The point where those two lines cross

This is a model, not a case study.

Put your own four inputs in before you quote the output to anyone who signs things.

Three things I still get wrong

1

Tables in PDFs

I have no reliable general answer. I parse them per document type and it does not transfer.

2

Questions spanning ten documents

Retrieval returns five chunks.
"Summarise every exception across all policies" is not a retrieval question and I stop pretending it is.

3

Freshness at scale

Re-indexing a moving corpus without a stale window is still manual work in every system I have built.

If I do not know, I will say I do not know and come back to you with a precise answer.

Two things you keep, and one piece of homework

1

Keep the audit

The folder report from Round 1. Re-run it every time the corpus changes and diff the two.

2

Keep the twenty

Your question set. It is the only thing here that still works after the model you use gets replaced.

3

The homework

One question set. Run it this Friday against your live index, and send me the first row where the right document was in the corpus and retrieval missed it.